# Vo Van Tu Tai

**AI & LLM Application Engineer · RAG & Agentic Workflows · Loop Engineering**

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## PROFESSIONAL SUMMARY

Information Systems Engineer (IUH, GPA 3.26/4.0 · Completed all degree requirements · Available immediately for full-time) with hands-on expertise building production-ready **AI & LLM application features for E-commerce, MarTech, and Logistics platforms**. Creator of a live production AI Logistics Assistant deployed at **tracking.nexus-ex.site**, featuring **Live RAG** (Vector search, document parsing, grounded citations) combined with **Agent Tool Calling and Loop Engineering** (multi-turn conversational state loops that extract shipment entities, calculate real-time pricing quotes, and trigger courier dispatch API calls). Proficient in **Python (FastAPI, data parsing/embeddings)**, **Node.js/TypeScript**, RESTful API contracts, and relational databases (PostgreSQL, MySQL, Redis). Proven track record bridging engineering and non-technical stakeholders (Founders, Ops, Marketing) with deep commitment to prompt quality, guardrail enforcement, and token cost optimization in fast-paced product environments.

## CORE COMPETENCIES & TECHNICAL TOOLS

**LLM & Generative AI:** Structured Prompt Engineering (Role-based system prompts, Few-shot prompting, JSON output enforcement), Token Budget & Context Window Management, Output Evaluation & Guardrails (preventing hallucinations on terms, SLAs & damage claims).

**RAG & Loop Engineering:** Knowledge Base Processing (Markdown parsing, semantic chunking), Text Embeddings, Vector Storage & Similarity Retrieval (FAISS, Chroma, PostgreSQL pgvector), Multi-Turn Stateful Loops, ReAct Agent Pattern, Tool / Function Calling.

**Programming & Backend:** Python (proficient in service integration, data pipelines, FastAPI), Node.js, TypeScript, PHP (OOP, MVC), RESTful API Design (OpenAPI 3.0), Webhooks, RabbitMQ Message Broker.

**Databases & DevOps:** PostgreSQL, MySQL (3NF & ACID transactions), Redis (In-memory caching, session store & distributed locks), Docker, Docker Compose, Git / GitHub, Postman, Linux / Bash scripting, Jira.

**Domain & Professional Skills:** Technical English comprehension (reading LLM papers & API specs), Vietnamese (Native), clear cross-functional communication with non-tech stakeholders, rapid adaptation in Agile startup sprints.

## PROFESSIONAL WORK EXPERIENCE

**HomeNest Software** — Ho Chi Minh City, Vietnam

**Business Analyst · PM Assistant (Engineering & Operations Liaison)** — *Mar 2026 – Sep 2026*

▸ **Project 1: Multi-Service On-Demand Utility Ecosystem & Dynamic Bidding**

- **Architecture & Multi-Service Scope Modeling:** Analyzed requirements for an integrated on-demand platform spanning home repairs, short-trip rides, interprovincial carpooling, and centralized Admin; authored unambiguous JSON data contracts and OpenAPI specs.
- **State-Machine & Interactive Bidding Logic:** Architected a two-way price bidding state machine (propose ↔ counter-bid ↔ accept/reject with TTL expiration); specified strict status locking and concurrency guards to prevent race conditions during checkout.
- **Third-Party API Cost & Battery Optimization:** Mitigated high third-party mapping API expenses and mobile battery drain by specifying an event-driven GPS checkpoint mechanism (triggering location updates strictly on order pickup/drop-off instead of continuous polling), saving substantial external API budget.
- **Technical Specification & Schema Integrity:** Authored PRDs, Use Cases, and User Stories (Given-When-Then); collaborated with Tech Lead and UI/UX to review wireframes against relational data models, eliminating schema mismatches before sprint kickoff.

▸ **Project 2: Commercial Marketplace & Cross-Border Delivery Coordination**

- **Backlog Ownership & Domain Mastery:** Stepped into an active development cycle; quickly audited backlogs and local compliance requirements to master listing lifecycles, content moderation, and courier workflows without disrupting delivery timelines.
- **Fulfillment, Order Cancellation & Dispute State Flow:** Partnered with clients on Change Requests; designed state machines and exception handling workflows for delivery tracking, automated order cancellations, and customer return/refund disputes.
- **Pragmatic Solution Thinking (International Lead Intake):** When the client requested urgent cross-border shipping capabilities, designed a lean conversational lead form for manual ops fulfillment rather than building heavy logistics software, achieving business goals while protecting **Scope Freeze**.
- **Go-Live Preparation, Functional Testing & UAT:** Authored Go-Live readiness checklists (App Store/Google Play policies); conducted hands-on Functional and Smoke testing against User Stories, triaged Jira defects, and coordinated successful client UAT sign-off.

## WORK EXPERIENCE (CONTINUED)

**J&T Express** Ho Chi Minh City, Vietnam

**Operations Process Coordinator (Full-time)** *Sep 2025 – Feb 2026*

- **SQL Data Reconciliation & System Integrity:** Formulated daily SQL queries to cross-verify discrepancies between physical warehouse manifests and core enterprise logistics systems, delivering data integrity reports to upper management.
- **Logistics Throughput Optimization:** Identified processing bottlenecks handling thousands of daily parcels; restructured shift workflows into an efficient unidirectional sorting pipeline (Intake → Weigh → Scan → Dispatch), substantially accelerating throughput.
- **Standard Operating Procedures (SOP):** Authored standard exception procedures categorizing delivery anomalies (damaged parcels, failed deliveries, lost items, COD claims) — providing foundational domain knowledge for downstream RAG chatbots and operational guardrails.

## FEATURED AI & TECHNICAL PROJECTS

**Nexus Express — Logistics AI Assistant, Live RAG & Loop Engineering** [Live Demo ↗] [GitHub ↗] AI & Logistics Project

*Python, FastAPI, Live RAG, Embeddings, Loop Engineering, Agent Tool Calling, React, NestJS, Docker* *Oct 2025 – Mar 2026*

- **Live RAG Architecture & Grounded Citations:** Engineered an end-to-end knowledge parsing and vector retrieval pipeline for operational policies, insurance terms, and parcel damage settlement rules (`02-insurance-and-claim-policy.md`); enforced transparent citation outputs, eliminating hallucinations regarding compensation limits and SLAs.
- **Loop Engineering (Multi-Turn Conversational State Machine):** Built an autonomous multi-turn intake loop: detects shipping intent → extracts structured entities (addresses, phone numbers, weight, COD amount) → prompts user for missing fields → triggers pricing quote tool → enters payment confirmation loop (Human-in-the-Loop) → locks order and provisions official tracking code.
- **Agent Tool Calling & UI Cards:** Implemented live backend function integration tools: `createRetailOrder` (auto-dispatch courier), `quoteShipmentOrder` (real-time distance/weight rate calculator), `trackClaimStatus` (insurance status tracking), and `getDamageAndClaimPolicy`; surfaced interactive Shipment Cards and contextual Action Chips in UI.
- **Gateway API Fallback & Token Budget:** Integrated REST API endpoints (`/api/v1/ai-assistant/message` and `/api/v1/chat/message`) with timeout guards (15s–30s) and internal heuristic fallback parsing, guaranteeing 24/7 service availability; optimized Context Window token budget via sessionStorage conversational history truncation.

**Tourist Accommodation Booking & Hospitality Platform** [GitHub ↗] Web Application Project

*Concurrency Control, RBAC, VNPay IPN Webhook, MySQL, Express.js, Docker* *Oct 2025 – Dec 2025*

- **Inventory Lock & Overbooking Defense (TTL Hold):** Solved critical room reservation race conditions during peak traffic spikes through a stateful two-phase booking hold: temporarily locks rooms in `Pending` state with a 15-minute TTL hold, auto-releasing inventory via timeout daemon if unconfirmed.
- **Multi-Tenant RBAC Architecture:** Architected an end-to-end booking platform with granular RBAC strictly isolating Guests, Hosts, and Super-Admins; protected sensitive transactional endpoints with middleware guards.
- **Cryptographic Payment Gateway:** Integrated VNPay payment gateway using HMAC-SHA512 checksum validation with idempotent IPN webhook processing; maintained sub-second query latencies under concurrent lookups.

**Song Tai Shop — Full-Stack E-Commerce Platform** [GitHub ↗] E-Commerce Platform

*PHP (OOP & MVC Architecture), MySQL (3NF & ACID), RESTful APIs, JavaScript (Fetch/AJAX), Docker Compose* *Nov 2025 – Jan 2026*

- **E-Commerce Architecture & ACID Concurrency:** Developed a full-stack online store with custom MVC architecture; protected inventory consistency during simultaneous checkout spikes using MySQL ACID transactions (`START TRANSACTION`, `COMMIT`, `ROLLBACK`) và pessimistic row locking (`SELECT ... FOR UPDATE`).
- **RESTful Catalog APIs & Conversational Commerce:** Designed standardized endpoints for product catalogs, SKU variants, cart management, and order states; understood end-to-end E-commerce customer lifecycles for MarTech automation.

## EDUCATION & ACADEMIC BACKGROUND

**Industrial University of Ho Chi Minh City (IUH)** Ho Chi Minh City, Vietnam

**Bachelor of Engineering (B.Eng.) in Information Systems · GPA: 3.26 / 4.0** *Sep 2022 – 2026 (Completed All Degree Requirements · Available Full-Time)*

- **Key Relevant Coursework:** Artificial Intelligence & Intelligent Systems Concepts, Software Engineering, Data Structures & Algorithms, Advanced Database Systems (SQL Optimization & Indexing), Service-Oriented Architecture (SOA), Systems Analysis & Design.